QUIZ 3 -MTH301-page1-4

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

10 / 12 pts

Question 1

Q1 4 / 6 pts

+ 0 pts Completely Incorrect

+ 6 pts Completely correct

✔ **+ 2 pts** Part (a) is correct.

✔ **+ 2 pts** Finding correct fixed point (0,0) in part (b).

+ 2 pts Proof of strict contraction is correct in part (b).

+ 4 pts Part (b) is correct.

1 How?

Question 2

Q2

Resolved **6** / 6 pts

2 (a)

+ 0 pts 2 (a) Completely Incorrect

✔ **+ 4 pts** 2(a) Completely Correct

+ 1 pt 2 (a) Partially Correct

+ 2 pts 2 (a) Partially Correct

+ 3 pts 2 (a) Partially Correct

2 (b)

+ 0 pts 2 (b) Completely Incorrect

✔ **+ 2 pts** 2 (b) Completely Correct

+ 1 pt 2 (b) Partially Correct

Regrade Request **Submitted on: Nov 05**

In part a) one mark was cut because of x-1/n^2 > x-1/N^2 because n>N want specified, but it was actually outside the box that gradescope automatically makes , so it wasn't visible to the grader. I am attaching the Google drive link showing that it is indeed written, i hope the request is considered

https://drive.google.com/file/d/1G29byvNgX7u_3dyaD9r9xHJiB-kcKfle/view?usp=drivesdk

Pls open and see the fact that I have written n>Ne just below the underlined statement

Done.

Reviewed on: Nov 09

ANALYSIS - I : MTH 301 : QUIZ : 2025-26

Name: YATIN PREETAM BHOJWANI Roll No: 241211

- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 25 Time: 18:30 - 19:45 .

1. (a) Find an example of a space M and a *strict contraction* map $F : M \to M$ having no fixed point where M is *not* complete.

   (b) Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be a map defined for any $(x_1, x_2) \in \mathbb{R}^2$ by $F(x_1, x_2) = \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$. Show that F is a *strict contraction* and find the unique fixed point. [2+4]

(a) ~~F: $M = [0,1]$~~ $(0,1]$ is an incomplete space and $f(x) = \frac{x}{2}$.

also $|f(y) - f(x)| = \left|\frac{y}{2} - \frac{x}{2}\right|$ $\le \frac{1}{2}|y-x|$

(a) take $M = (0,1]$

this is incomplete space

$\{x_n : \frac{1}{n}$ is cauchy but does not converge$\}$

Let $F : M \to M$

$F(x) = \frac{x}{2}$

show F is strict contraction

$|F(y) - F(x)| = \left|\frac{y}{2} - \frac{x}{2}\right| = \frac{1}{2}|y-x|$

$\le \frac{1}{2}|y-x|$

$\Rightarrow |F(y) - F(x)| \le \frac{1}{2}|y-x|$

(b) $F : \mathbb{R}^2 \to \mathbb{R}^2$

$F(x_1, x_2) = \frac{1}{4}(x_1 + x_2,\ x_1 + 2x_2)$

~~take 2~~

take $y, x \in \mathbb{R}^2$

$y = (y_1, y_2)$, $x = (x_1, x_2)$

~~to show it is~~

$\|F(y_1, y_2) - F(x_1, x_2)\|$

$= \left|\frac{1}{4}(y_1 + y_2\right.$

to show it is contraction

~~$d(y,x)$~~ $d(f(y), f(x)) \le \alpha\, d(y,x)$

here d is the $\|\cdot\|$ norm

$\left\|\frac{1}{4}(y_1 + y_2, y_1 + 2y_2) - \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)\right\|$

$= \frac{1}{4}\|(y_1 + y_2 - x_1 - x_2,\ y_1 + 2y_2 - x_1 - 2x_2)\|$

$= \frac{1}{4}\sqrt{(y_1 + y_2 - x_1 - x_2)^2 + (y_1 + 2y_2 - x_1 \ldots}$

~~$= \frac{1}{4}\sqrt{y_1^2 + y_2^2}$~~

$= \frac{1}{4}\sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + 2(y_1 - x_1)(y_2 - x_2) + (y_1 - x_1)^2 + (2y_2 - 2x_2)^2 + 2(y_1 - x_1)(2y_2 - 2x_2)}$

$= \frac{1}{4}\sqrt{2(y_1 - x_1)^2 + 5(y_2 - x_2)^2 + 6(y_1 - x_1)(y_2 - x_2)}$

~~$= \frac{1}{4}\sqrt{(y_1 - x_1)^2 + \frac{5}{2}(y_2 - x_2)^2 + 3(y_1 - x_1)(y_2 - x_2)}$~~

$= \frac{1}{16}$

$= \sqrt{\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2)}$

$\le \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} = \|y - x\|$

[as $\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2) < (y_1 - x_1)^2 + (y_2 - x_2)^2$]

$\Rightarrow \|f(y) - f(x)\| < \|y - x\|$

hence a strict contraction.

finding fixed pt. (unique)

$F(x_1, x_2) = (x_1, x_2)$

$\frac{1}{4}(x_1 + x_2,\ x_1 + 2x_2) = (x_1, x_2)$

$4x_1 = x_1 + x_2$, $4x_2 = x_1 + 2x_2$ $\Rightarrow x_1 = 0$, $x_2 = 0$

$\boxed{\text{fixed pt is } (0,0)}$

.... Rough Work ....

fixed pt. strict contraction mapping & complete space.

if $d|f(x) - f(y)| < \alpha d(x-y)$

if i take x_0

$(f^n(x_0))$ is cauchy

$x \Rightarrow \lim_{n\to\infty} f^n(x_0)$.

$3x_1 = x_2$.

$\frac{1}{4}(x_1 + x_2,\ x_1 + 2x_2)$,

$2[y_1^2 + x_1^2 - 2y_1x_1]$

$+5[y_2^2 + x_2^2 - 2y_2x_2]$

$+6y_1y_2 - 6y_1x_2 - 6x_1y_2 + 6x_1x_2$

$2y_1^2 + 2x_1^2 - 4y_1x_1$

$+5y_2^2 + 5x_2^2 - 10y_2x_2$

$+6y_1y_2 - 6y_1x_2$

eqn : $x = \frac{\pi}{2}$ has no roots in $[0,1]$

$-2x_2)^2$

1

4

$= (x_1, x_2)$

$\frac{1}{4}(x_1 + 2x_1) = x_2$

$x_1 + 2x_2 = 4$

$\frac{1}{4}(x_1 + x_2) = x_{12}$

$x_1 + x_2 = 4x_1$

$-6x_1 y_2$

$+6x_1 x_2.$

$2y_1^2 + 5y_2^2 + 2x_1^2$

$+5x_2^2 - 10y_1$

2. (a) Let $g : \mathbb{R} \to \mathbb{R}$ be uniformly continuous, and define $f_n(x) := g\left(x - \frac{1}{n^2}\right)$. Show that f_n uniformly converges to g on $\mathbb{R}$.

(b) Let $f : [0, 3] \to \mathbb{R}$ be a continuous function. Justify whether $\lim_{n\to\infty} \int_{\frac{1}{n}}^{2+\frac{1}{n}} f(y)dy = \int_0^2 f(y)dy$.

[4+2]

(a) $g : \mathbb{R} \to \mathbb{R}$. is uniformly continuous.

$\forall \varepsilon > 0$, $\exists \delta_\varepsilon > 0$ s.t $|y - x| < \delta_\varepsilon$

$\Rightarrow |g(y) - g(x)| < \varepsilon$

and that $x - \frac{1}{n^2} < x + \frac{1}{N_\varepsilon^2}$

$\forall n > N_\varepsilon$

(b) ~~$\lim_{n\to\infty} \int_{\frac{1}{n}}$~~ as f is continuous integral exists.

By Fundamental Theorem of Calculus. (Riemman integral

$\int_a^b f(y)\, dy = F(b) - F(a)$

$f(x)$ is continuous

Yes it is true.